Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-J Vascular Neoplasms

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-10-C01				
NPLEX-P1-CARD-18-10-C02				
NPLEX-P1-CARD-18-10-C03				
NPLEX-P1-CARD-18-10-C04				
NPLEX-P1-CARD-18-10-C05				
NPLEX-P1-CARD-18-10-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C01-BD1

Hemangioma against malformation, and the life of an infantile hemangioma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-BD1 Date: _____

Without looking at your notes, write a definition of a hemangioma and a definition of a vascular malformation that make clear which one involves proliferating endothelial cells. Underneath, draw the life cycle of an infantile hemangioma as a timeline with five stages, from what is visible at birth through to what replaces the tumor at the end, and write beside each stage what the endothelial cells are doing. Add the incidence in infants, the groups in which it occurs more often, the color of a superficial lesion and of a deep one, and the marker its endothelium expresses in every phase. Then write what a capillary hemangioma looks like down the microscope and what a cavernous hemangioma looks like, one line each, and state which of the two regresses and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C01-BD2

Site, growth rate and the harm they cause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-BD2 Date: _____

Four infants have hemangiomas: one on the upper eyelid covering the pupil, one spread across the beard area of the jaw and neck, one large and segmental across half the face, and one large lesion in the liver. For each infant write what harm you would expect and the mechanism that produces it, including the enzyme and the hormone involved in the hepatic lesion. Then write which complication is the most common of all and explain it from the growth rate of the tumor and the blood supply of the skin above it. Next, a newborn has a red facial patch that was fully formed at birth and has since grown only in proportion to the child. Write what this lesion is, whether its endothelium would stain for GLUT1, whether you expect it to regress, and the syndrome and somatic mutation it is linked to. Finish by writing which two lesions cause the platelet-trapping consumptive coagulopathy and why the ordinary infantile hemangioma does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-10-C01-RT01

Rebuild the life cycle of an infantile hemangioma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C01 Activity: NPLEX-P1-CARD-18-10-C01-RT01 Date: _____

Rebuild the five-step life cycle of an infantile hemangioma from memory, one step per line and in order, from the appearance at birth to the final state. Beside the proliferation step write the marker the endothelium carries and the other tissue in the body known for that marker. Beside the regression step write the form of cell death involved and the tissue that takes the place of the vessels, and add the two changes that can be left behind in the skin. Underneath the steps write the one feature of this sequence that separates a tumor from a malformation.

Rebuild from memory: Life cycle of an infantile hemangioma

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-10-C01-CP01

Infantile hemangioma, cavernous hemangioma and port-wine stain

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Infantile (strawberry) hemangioma	Cavernous hemangioma	Port-wine stain
What is visible at birth			
Pattern and speed of growth			
Does it regress			
GLUT1 staining			
What you see on histology			
Usual sites			
The association most often tested			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C02-BD1

Three red papules that are not hemangiomas of infancy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C02 Activity: NPLEX-P1-CARD-18-10-C02-BD1 Date: _____

For pyogenic granuloma, cherry hemangioma and bacillary angiomatosis, write a block of four lines each: the patient and setting in which it appears, what the lesion looks like and how fast it grows, what the histology shows including every cell type and any organism, and what happens to it over time. Write the alternative histologic name of pyogenic granuloma and explain which two words in the name pyogenic granuloma are inaccurate and why. Then write what the pregnancy form is called, where in the mouth it sits, and the two hormones that amplify the response along with the local factor they amplify it against. For bacillary angiomatosis, name both organisms and how each reaches the patient, and write the transcription factor and growth factor through which the bacteria drive capillary growth, along with the stain that shows the organisms.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C02-BD2

A purple papule in a man with a CD4 count of 40

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C02 Activity: NPLEX-P1-CARD-18-10-C02-BD2 Date: _____

A man with untreated HIV infection and a CD4 count of 40 cells per microliter has several violaceous papules on his arms, a fever of 38.4 degrees C (101.1 degrees F), and a cat at home. Write the two diagnoses that fit both the skin finding and the immune state, and write why the color of the lesions does not separate them. Then write what on the biopsy would settle it in favour of each one, listing the shape of the vessels, the lining cells, the inflammatory cells and any nuclear or silver stain result for both. Write which clinical feature in this man leans towards an infection and why. Finish by writing what happens to each lesion over the following months if the organism is cleared in one case and if the CD4 count recovers in the other, and explain the difference from what sustains each proliferation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-10-C02-CP02

Red and purple vascular lesions side by side

Fill in the table from memory for all five lesions, then check it against the notes.

Lesion	Typical patient	Cause	Key histology	Course
Infantile hemangioma				
Pyogenic granuloma				
Cherry hemangioma				
Bacillary angiomatosis				
Kaposi sarcoma				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C03-BD1

How HHV-8 drives the spindle cell

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-BD1 Date: _____

Write what kind of virus HHV-8 is, how it spreads from person to person, and the one sentence that explains why most infected people never develop the tumor. Name the tumor cell, the lineage it comes from, and four markers it carries. Then draw the latent programme: for the viral cyclin, for LANA and for viral FLIP, write which host protein or pathway each one acts on, what that action does to the cell cycle or to cell death, and which checkpoint is lost. Beside the viral cyclin write the host protein it is a homolog of, the CDK it binds, the inhibitors it resists, the protein it causes to be phosphorylated and the transcription factor that is then freed. Separately write what the small lytic fraction of cells secretes or expresses, the growth factor that rises as a result, and what those signals do to neighbouring cells. Finish with one line on why early lesions are polyclonal and advanced nodules may be clonal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C03-BD2

From slit-like spaces to the findings at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-BD2 Date: _____

Build the three stages of a Kaposi sarcoma lesion from memory as three rows, writing for each what the lesion looks like to the eye and what the microscope shows. Then explain each of the following findings from the structure you have just described, two sentences each: the purple color of a new lesion, the brown color of an old one, swelling of the whole limb, and bleeding from a lesion in the gut. Write which nuclear stain identifies the virus in the spindle cells and at which stages it works. Next, a transplant recipient on T-cell immunosuppression develops widespread lesions and the immunosuppression is then lowered. Write what you expect the lesions to do and justify it from both the latent viral genes and the cells that normally police infected cells. Finish by naming the two other diseases HHV-8 causes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-10-C03-RT01

Rebuild the path from infection to a nodule

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C03 Activity: NPLEX-P1-CARD-18-10-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from HHV-8 infecting an endothelial-lineage cell to a Kaposi sarcoma nodule, one step per line and in order. Mark which steps belong to the latent programme, which belong to the lytic programme and which belong to the host immune system. Beside the latent steps write the host proteins put out of action. Beside the lytic step write the two signals released and the growth factor that rises. Beside the immune step write the two cell types whose loss of control permits expansion. Underneath, write one line on what each of the three groups contributes, so that removing any one of them would stall the sequence.

Rebuild from memory: From HHV-8 infection to a Kaposi sarcoma nodule

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-10-C03-PR01

Predict what happens when single viral genes are lost

A laboratory studies HHV-8 strains in which one viral gene at a time has been inactivated in infected endothelial-lineage cells. For each item below write the direction of the change you expect in the infected cell and give the host protein or pathway that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
With the viral cyclin inactivated, what happens to Rb phosphorylation and to the G1 to S transition		
With the viral cyclin inactivated, what happens to the braking effect of p16, p21 and p27 on the cell cycle		
With LANA lost, what happens to the viral episome in daughter cells after division		
With LANA lost, what happens to p53 and Rb activity and to the fate of a cell that is dividing without a signal to do so		
With viral FLIP lost, what happens to NF-kappaB signalling and to apoptosis		
With the lytic genes silenced in the whole lesion, what happens to viral IL-6 and VEGF levels and to the growth of neighbouring cells		
With cytotoxic T-cell and NK cell function restored in the host while all viral genes remain intact, what happens to the size of the lesions		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C04-BD1

Build the four forms of Kaposi sarcoma from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C04 Activity: NPLEX-P1-CARD-18-10-C04-BD1 Date: _____

On blank paper build a table of the four clinical forms of Kaposi sarcoma with one row each for classic, endemic African, transplant-associated and AIDS-associated. Use the columns population affected, typical sites and course, and fill every cell, including ages, ancestry, HIV status and the route by which the virus is usually acquired where the notes give it. Above the table write the one feature every form shares, both in terms of the virus and in terms of the histology. Underneath the table write the two host variables that decide how the tumor behaves, and give an example of what mild immune decline produces and what profound suppression produces.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C04-BD2

Four patients, four courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C04 Activity: NPLEX-P1-CARD-18-10-C04-BD2 Date: _____

Four patients have biopsy-proven Kaposi sarcoma: a 74-year-old HIV-negative man of Mediterranean descent with plaques on both feet; a 6-year-old in sub-Saharan Africa with large cervical and axillary lymph nodes and no skin lesions; a woman 14 months after kidney transplantation with lesions on the skin, palate and in the stomach; and a man with HIV infection with lesions on the face, hard palate and in the lungs. For each patient name the form, write the cofactor that permitted the tumor, and predict the pace and extent of the disease over the next year. Then write which patient is most likely to die of something unrelated and why, which patient has the illness that defines AIDS, and which patient's disease is most likely to shrink without any treatment aimed at the tumor itself, explaining each from immune status. Finish by writing why the AIDS form is far more common in men who have sex with men than in other groups with HIV, and what you expect the lesions to do transiently during early immune recovery and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C05-BD1

Myxoma and rhabdomyoma from the ground up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C05 Activity: NPLEX-P1-CARD-18-10-C05-BD1 Date: _____

Write a block on myxoma and a block on rhabdomyoma. In each block give the age group it belongs to, any sex difference, the chamber and exact site it arises from with the proportions the notes give, whether it is a true neoplasm or a hamartoma, the cell of origin, and the histology including the matrix or the stored substance and the name given to the characteristic cell. Add for myxoma the range of gross appearances from firm to gelatinous, and for rhabdomyoma whether the lesions are single or multiple and what happens to them in early childhood. Then write the inherited syndrome linked to each, with the gene, the protein it encodes, the pathway involved and the pattern of inheritance, and for the myxoma syndrome add the two non-cardiac findings. Finish with one line on how common primary cardiac tumors are compared with metastases to the heart.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C05-BD2

Why a benign tumor makes a patient faint

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C05 Activity: NPLEX-P1-CARD-18-10-C05-BD2 Date: _____

Draw a left atrium with a stalked mass on the interatrial septum and rebuild in six steps how it produces symptoms, from where it is attached through to what breaks off, one step per line. Then write the three groups of findings a myxoma produces. For the obstructive group write the valve lesion it imitates, the added sound and when in the cycle it occurs, and why the murmur and the symptoms change with body position. For the embolic group write where the fragments go from a left atrial tumor and where they go from a right atrial tumor, with the consequence in each case. For the constitutional group write the cytokine, the response it drives in the liver, the two laboratory values that rise and the reason for the anemia. Finish by writing the two ways a rhabdomyoma in a child causes trouble, and explain from the position of the tumor in the heart why its problems differ from those of a myxoma.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-10-C05-CA01

Breathlessness, fainting on standing and a high ESR

A 45-year-old woman comes to the clinic because of 4 months of breathlessness on exertion and two fainting spells, both of which happened as she stood up from a chair. She has lost 5 kg (11 lb) over the same period and reports low-grade fevers and aching in her hands. Temperature is 37.8 degrees C (100.0 degrees F), pulse is 96/min and regular, respirations are 20/min, and blood pressure is 112/70 mm Hg. On auscultation a low-pitched sound follows the second heart sound, and a diastolic murmur is heard at the apex whose loudness changes when she lies on her left side. Hemoglobin is 10.4 g/dL, the erythrocyte sedimentation rate is 78 mm/h, and C-reactive protein is raised. Blood cultures are repeatedly negative, and there is no history of rheumatic fever.

1. Recognize. Name the lesion, the chamber it sits in and the structure within that chamber it is usually attached to, and name the sound that follows the second heart sound.

2. Explain. Explain the mechanism that links the lesion to her breathlessness and to fainting that happens only when she changes position, following the pressures from the chamber of origin back into the lungs.

3. Differentiate. Write how this picture differs from rheumatic mitral stenosis and from infective endocarditis, naming for each the findings in this woman that do not fit it.

4. Investigate. Write which single imaging study you would use, what it would show and where, and name the two laboratory findings already present that support the systemic part of the picture.

5. Decide. State which complication of this lesion would be the most damaging if a fragment broke away, and name the organ most likely to be affected from a tumor in this chamber.

6. Justify. Justify your account of the fevers, the weight loss, the raised sedimentation rate and the anemia from a single substance the tumor secretes, and explain why negative blood cultures do not argue against the diagnosis.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-10-C06-BD1

Angiosarcoma: what it is, where it sits, what it looks like

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C06 Activity: NPLEX-P1-CARD-18-10-C06-BD1 Date: _____

Write what cell angiosarcoma arises from, the age group it belongs to, and the sites where it most often appears, including the chamber of the heart and the skin site typical of elderly patients. Describe the histology of a well-differentiated tumor in two or three sentences, naming what the channels do to the surrounding collagen and what the lining cells look like, then describe a poorly differentiated tumor and write the two markers used to prove the lineage when the cells no longer form channels. Underneath, list all six entries from the risk factor table with the setting and the main site for each. Finish by writing the behavior of the tumor: how it spreads at its margin, whether it is single or multifocal in skin, where it metastasizes, the five-year survival figure from the notes and which site does worse.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-10-C06-BD2

Same tumor, two different organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-J Vascular Neoplasms Competency: NPLEX-P1-CARD-18-10-C06 Activity: NPLEX-P1-CARD-18-10-C06-BD2 Date: _____

A 68-year-old man who spent 20 years in a plant making polyvinyl chloride has a liver mass. A 61-year-old woman who had a radical mastectomy with axillary node dissection 12 years ago, and whose arm has been swollen ever since, has a purple mass in that arm. Write the tumor both have, then explain from first principles why the organ is different in each case. For the man, rebuild the five-step sequence from inhaled monomer to malignant proliferation, naming the enzyme, the reactive product and what that product does to DNA, and write why the liver rather than the lung is the target. For the woman, name the syndrome and write the two features of a chronically lymphedematous limb that are thought to allow the tumor to arise there. Then add a third patient who had chest wall irradiation 9 years ago and now has a mass in the treated field: write the genetic change these tumors often carry. Finish by writing the one property these exposures share that explains why all three patients present so long after the exposure, and the two other hepatic exposures from the notes with what each one is and how it reached the liver.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-10-C01-Q01

A red lesion that appeared after birth

A 5-week-old girl born at 33 weeks of gestation is brought to the physician because of a bright red raised lesion on her cheek. Her mother says the skin there looked faintly pale at birth, that a small red spot appeared at 2 weeks of age, and that it has enlarged steadily since. Biopsy of a similar lesion in another infant shows lobules of closely packed thin-walled capillaries whose endothelium stains for GLUT1. Which of the following is the most likely course of this lesion?

A. Rapid growth for several months, then apoptosis of the endothelium with fibrofatty replacement

B. Steady enlargement in proportion to the child, with darkening and thickening in adult life

C. Progressive dilatation of large thin-walled channels with thrombosis and calcification

D. Continued growth with infiltration of deep tissue and metastasis to the lungs

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C02-Q02

Purple nodules in an immunocompromised host

A 36-year-old man with untreated HIV infection and a CD4 count of 35 cells per microliter comes to the clinic because of fever and several red to purple nodules on his trunk that appeared over 3 weeks. He adopted two kittens 2 months ago. Biopsy of a nodule shows a proliferation of rounded capillaries lined by plump epithelioid endothelial cells, a dense neutrophilic infiltrate with nuclear debris, and clumps of granular purple material that resolve into bacilli on Warthin-Starry staining. Which of the following best explains the vascular proliferation in this lesion?

A. A viral cyclin homolog that binds CDK6 and phosphorylates Rb in infected spindle cells

B. A somatic activating mutation in GNAQ in dermal endothelial cells

C. Bacterial induction of hypoxia-inducible factor 1 alpha with a rise in VEGF

D. Endothelial expression of type 3 iodothyronine deiodinase with local hormone inactivation

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C03-Q03

The gene that removes two checkpoints at once

A 52-year-old man who received a kidney transplant 18 months ago and takes T-cell immunosuppression has violaceous plaques on his legs and trunk. Biopsy shows fascicles of spindle cells surrounding slit-like vascular spaces containing extravasated erythrocytes, with scattered lymphocytes, plasma cells and hemosiderin-laden macrophages. Immunostaining shows a nuclear antigen in the spindle cells. The protein detected by that stain tethers the viral episome to host chromosomes during mitosis. Which additional effect does this same protein have on the infected cell?

A. It activates NF-kappaB and thereby blocks apoptosis

B. It binds and inhibits both p53 and Rb, removing the checkpoint response to unscheduled proliferation

C. It acts as a constitutively active G protein-coupled receptor that raises VEGF

D. It resists the cyclin-dependent kinase inhibitors p16, p21 and p27

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-10-C06-Q04

A liver mass decades after an industrial exposure

A 70-year-old man who worked for 22 years in a plant that polymerized vinyl chloride into plastic piping has right upper quadrant pain and weight loss. Imaging shows a large vascular mass in the right lobe of the liver, and biopsy shows irregular anastomosing channels dissecting between collagen bundles and lined by plump, multilayered atypical cells that stain for CD31. Which of the following best explains why the liver is the organ affected in this man?

A. Hepatic macrophages retained the inhaled particles and irradiated adjacent cells with alpha particles for decades

B. Impaired lymphatic drainage of the liver reduced local immune surveillance in a growth factor rich tissue

C. Amplification of MYC in hepatic endothelium followed the prolonged exposure

D. Hepatic cytochrome P450 2E1 oxidized the monomer to a reactive epoxide that formed DNA adducts in sinusoidal cells

Why did you pick it?